Microsoft Defender Repair Script

Simulation Results & End User Impact Analysis

Script Version: 1.1
Date: January 6, 2026
Author: Brad Brown
Developed with: Claude Sonnet 4.5

Executive Summary

This document provides simulation results and end-user impact analysis for the Microsoft Defender Repair script deployed via NinjaRMM. The script offers six repair options with varying levels of intervention, from passive diagnostics to forced repair sequences.

Key Features

- ☑ Multiple repair strategies with intelligent decision-making
- ☑ Minimal end-user disruption (most operations are invisible)
- ☑ Comprehensive logging for troubleshooting
- ☑ Safe by default with escalation options
- ☑ Third-party antivirus detection and guidance

Repair Options Overview

Option	Purpose	User Impact	Duration	Recommended Use
Diagnostic Check Only	Status validation	None	6-8s	Initial assessment
Enable Defender	Activate protection	Minimal	5-10s	Post-AV removal
Reset to Default	Clear policy conflicts	Moderate	10-15s	Policy issues
Remove Third-Party AV	Detection & guidance	None	5-8s	AV conflicts
Full Repair (Smart)	Intelligent repair	None to Minimal	8-20s	**Primary option**
Force Full Repair	Nuclear option	Moderate	18-25s	Last resort

Simulation Scenarios

Scenario 1: Full Repair (Intelligent) - Healthy System

System State:

- Microsoft Defender: ☑ Running
- Real-Time Protection: ☑ Enabled

- Third-Party AV: ❌ None detected

Console Output:

```
[2026-01-06 15:15:22] [INFO] === Microsoft Defender Repair Script Started ===
[2026-01-06 15:15:22] [INFO] Selected Action: Full Repair (All Steps)
[2026-01-06 15:15:22] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:15:23] [INFO] Initial Status - Service Running: True
[2026-01-06 15:15:23] [INFO] Initial Status - Real-Time Protection: True
[2026-01-06 15:15:23] [INFO] Initial Status - Antivirus Enabled: True
[2026-01-06 15:15:23] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:15:24] [INFO] Performing intelligent full repair sequence...
[2026-01-06 15:15:24] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:15:25] [INFO] No third-party antivirus detected
[2026-01-06 15:15:25] [INFO] No third-party AV detected - skipping removal step
[2026-01-06 15:15:25] [INFO] Defender settings appear normal - skipping reset
[2026-01-06 15:15:25] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:15:26] [INFO] Defender already enabled and running - skipping
enable step
[2026-01-06 15:15:29] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:15:30] [INFO] === Final Status ===
[2026-01-06 15:15:30] [INFO] Service Running: True
[2026-01-06 15:15:30] [INFO] Real-Time Protection: True
[2026-01-06 15:15:30] [INFO] Antivirus Enabled: True
[2026-01-06 15:15:30] [INFO] ✓ Microsoft Defender is now active
[2026-01-06 15:15:30] [INFO] === Script Completed ===
```

End User Experience:

Metric	Value
Duration	8 seconds
Screen Impact	None - completely invisible
Performance Impact	Minimal - status checks only
Interruption Level	**Zero**
User Notifications	None
Service Interruption	No
Result	☑ Validation only, no changes needed

Outcome: Script intelligently detected healthy system and skipped all repair steps.

Scenario 2: Full Repair (Intelligent) - Broken System

System State:

- Microsoft Defender: ⚠ Service running but disabled

- Real-Time Protection: ❌ Disabled
- Third-Party AV: ⚠️ AVG AntiVirus detected

Console Output:

```
[2026-01-06 15:20:45] [INFO] === Microsoft Defender Repair Script Started ===
[2026-01-06 15:20:45] [INFO] Selected Action: Full Repair (All Steps)
[2026-01-06 15:20:45] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:20:46] [INFO] Initial Status - Service Running: True
[2026-01-06 15:20:46] [INFO] Initial Status - Real-Time Protection: False
[2026-01-06 15:20:46] [INFO] Initial Status - Antivirus Enabled: False
[2026-01-06 15:20:46] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:20:47] [WARNING] Third-party AV detected: AVG AntiVirus Free
[2026-01-06 15:20:47] [INFO] Performing intelligent full repair sequence...
[2026-01-06 15:20:47] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:20:48] [INFO] Third-party AV detected - attempting removal...
[2026-01-06 15:20:48] [INFO] Attempting to remove third-party antivirus...
[2026-01-06 15:20:48] [INFO] Found: AVG AntiVirus Free
[2026-01-06 15:20:49] [INFO] Found AV program: AVG AntiVirus Free
[2026-01-06 15:20:49] [INFO] Uninstall string: C:\Program Files\AVG\AVG PC
TuneUp\TUInstallHelper.exe
[2026-01-06 15:20:49] [WARNING] Please manually uninstall or use vendor-specific
removal tool
[2026-01-06 15:20:49] [INFO] Third-party AV check complete. Manual removal may be
required.
[2026-01-06 15:20:51] [INFO] Defender policy overrides detected - reset needed
[2026-01-06 15:20:51] [INFO] Performing reset to default settings...
[2026-01-06 15:20:51] [INFO] Resetting Microsoft Defender to default settings...
[2026-01-06 15:20:51] [INFO] Removing policy overrides...
[2026-01-06 15:20:51] [INFO] Removed policy:
HKLM:\SOFTWARE\Policies\Microsoft\Windows Defender
[2026-01-06 15:20:52] [INFO] Resetting Defender preferences...
[2026-01-06 15:20:53] [INFO] Restarting Windows Defender service...
[2026-01-06 15:20:58] [INFO] Defender reset to default settings successfully
[2026-01-06 15:21:00] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:21:01] [INFO] Defender needs to be enabled...
[2026-01-06 15:21:01] [INFO] Enabling Microsoft Defender...
[2026-01-06 15:21:01] [INFO] Enabling Real-Time Protection via registry...
[2026-01-06 15:21:02] [INFO] Microsoft Defender enabled successfully
[2026-01-06 15:21:05] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:21:06] [INFO] === Final Status ===
[2026-01-06 15:21:06] [INFO] Service Running: True
[2026-01-06 15:21:06] [INFO] Real-Time Protection: True
[2026-01-06 15:21:06] [INFO] Antivirus Enabled: True
[2026-01-06 15:21:06] [INFO] ✓ Microsoft Defender is now active
[2026-01-06 15:21:06] [INFO] === Script Completed ===
```

End User Experience:

Metric	Value

Metric	Value
Duration	21 seconds
Screen Impact	Windows Security icon briefly changes
Performance Impact	Brief CPU spike during service restart
Interruption Level	**Minimal (3-5 seconds)**
User Notifications	Brief toast: "Windows Security - Taking action..."
Service Interruption	Yes - 3-5 second gap during restart
Security Icon State	❌ → ⚠️ → ☑️

User-Visible Effects:

- 🛡️ Windows Security icon shows brief transition
- 📱 Possible notification: "Your device is protected"
- ⚡ Brief system tray activity

Outcome: ☑️ Successfully repaired Defender, identified third-party AV for manual removal.

Scenario 3: Force Full Repair (No Checks) - Appears Healthy

System State:

- Microsoft Defender: ☑️ Appears fully functional
- Real-Time Protection: ☑️ Enabled
- Issue: Hidden corruption or policy conflict

Console Output:

```
[2026-01-06 15:25:10] [INFO] === Microsoft Defender Repair Script Started ===
[2026-01-06 15:25:10] [INFO] Selected Action: Force Full Repair (No Checks)
[2026-01-06 15:25:10] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:25:11] [INFO] Initial Status - Service Running: True
[2026-01-06 15:25:11] [INFO] Initial Status - Real-Time Protection: True
[2026-01-06 15:25:11] [INFO] Initial Status - Antivirus Enabled: True
[2026-01-06 15:25:11] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:25:12] [INFO] Performing FORCED full repair sequence (no checks)...
[2026-01-06 15:25:12] [WARNING] WARNING: This will execute all repair steps
regardless of current state
[2026-01-06 15:25:12] [INFO] Attempting to remove third-party antivirus...
[2026-01-06 15:25:12] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:25:13] [INFO] No third-party antivirus detected
[2026-01-06 15:25:13] [INFO] Third-party AV check complete. Manual removal may be
required.
[2026-01-06 15:25:15] [INFO] Resetting Microsoft Defender to default settings...
[2026-01-06 15:25:15] [INFO] Removing policy overrides...
[2026-01-06 15:25:15] [INFO] Resetting Defender preferences...
[2026-01-06 15:25:16] [INFO] Restarting Windows Defender service...
```

```
[2026-01-06 15:25:21] [INFO] Defender reset to default settings successfully
[2026-01-06 15:25:23] [INFO] Enabling Microsoft Defender...
[2026-01-06 15:25:23] [INFO] Enabling Real-Time Protection via registry...
[2026-01-06 15:25:24] [INFO] Microsoft Defender enabled successfully
[2026-01-06 15:25:27] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:25:28] [INFO] === Final Status ===
[2026-01-06 15:25:28] [INFO] Service Running: True
[2026-01-06 15:25:28] [INFO] Real-Time Protection: True
[2026-01-06 15:25:28] [INFO] Antivirus Enabled: True
[2026-01-06 15:25:28] [INFO] ✓ Microsoft Defender is now active
[2026-01-06 15:25:28] [INFO] === Script Completed ===
```

End User Experience:

Metric	Value
Duration	18 seconds
Screen Impact	Security icon flickers during restart
Performance Impact	CPU spike, possible momentary lag
Interruption Level	**Moderate (5-8 seconds)**
User Notifications	"Windows Security - Restarting..." + "Your device is protected"
Service Interruption	Yes - protection down for ~5 seconds
Security Icon State	☑ → ⚠ (briefly) → ☑

⚠ **Security Window:** System unprotected for approximately 5 seconds during service restart.

When to Use:

- Smart repair completed but issues persist
- Subtle corruption not detected by status checks
- Escalation after other options fail

Outcome: ☑ Forced rebuild of all Defender components regardless of reported status.

Scenario 4: Force Full Repair - Severely Corrupted System

System State:

- Microsoft Defender: ❌ PowerShell module missing/corrupted
- Real-Time Protection: ❌ Cannot determine
- Issue: Deep Windows component corruption

Console Output:

```
[2026-01-06 15:30:05] [INFO] === Microsoft Defender Repair Script Started ===
[2026-01-06 15:30:05] [INFO] Selected Action: Force Full Repair (No Checks)
```

```
[2026-01-06 15:30:05] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:30:06] [ERROR] Error checking Defender status: The term 'Get-
MpComputerStatus' is not recognized
[2026-01-06 15:30:06] [WARNING] Unable to get initial Defender status
[2026-01-06 15:30:06] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:30:07] [INFO] Performing FORCED full repair sequence (no checks)...
[2026-01-06 15:30:07] [WARNING] WARNING: This will execute all repair steps
regardless of current state
[2026-01-06 15:30:07] [INFO] Attempting to remove third-party antivirus...
[2026-01-06 15:30:07] [INFO] Checking for third-party antivirus software...
[2026-01-06 15:30:08] [INFO] No third-party antivirus detected
[2026-01-06 15:30:08] [INFO] Third-party AV check complete. Manual removal may be
required.
[2026-01-06 15:30:10] [INFO] Resetting Microsoft Defender to default settings...
[2026-01-06 15:30:10] [INFO] Removing policy overrides...
[2026-01-06 15:30:10] [INFO] Removed policy:
HKLM:\SOFTWARE\Policies\Microsoft\Windows Defender
[2026-01-06 15:30:11] [INFO] Resetting Defender preferences...
[2026-01-06 15:30:11] [ERROR] Error resetting Defender: The term 'Set-
MpPreference' is not recognized
[2026-01-06 15:30:13] [INFO] Enabling Microsoft Defender...
[2026-01-06 15:30:13] [INFO] Starting Windows Defender service...
[2026-01-06 15:30:18] [INFO] Enabling Real-Time Protection via registry...
[2026-01-06 15:30:19] [INFO] Microsoft Defender enabled successfully
[2026-01-06 15:30:22] [INFO] Checking Microsoft Defender status...
[2026-01-06 15:30:22] [ERROR] Error checking Defender status: The term 'Get-
MpComputerStatus' is not recognized
[2026-01-06 15:30:22] [INFO] === Final Status ===
[2026-01-06 15:30:22] [WARNING] ⚠ Microsoft Defender may require additional
action or system reboot
[2026-01-06 15:30:22] [INFO] === Script Completed ===
```

End User Experience:

Metric	Value
Duration	17 seconds
Screen Impact	Security icon may remain in error state
Performance Impact	Normal
Interruption Level	Minimal
User Notifications	None (PowerShell module unavailable)
Service Interruption	Minimal
Security Icon State	Likely stays ⚠ or ❌

⚠ **Critical Issue:** PowerShell Defender module is missing or corrupted.

Follow-up Actions Required:

1. Run DISM: `DISM /Online /Cleanup-Image /RestoreHealth`
2. Run SFC: `sfc /scannow`
3. Consider Windows Repair/Reset
4. Check Windows Update status

Outcome: ⚠ Partial repair attempted. Registry keys set but cmdlets unavailable. Deeper Windows repair needed.

Diagnostic Check Only - Sample Output

Purpose: Non-invasive status validation before any repair attempts.

Console Output:

```
[2026-01-06 14:32:15] [INFO] === Microsoft Defender Repair Script Started ===
[2026-01-06 14:32:15] [INFO] Selected Action: Diagnostic Check Only
[2026-01-06 14:32:15] [INFO] Checking Microsoft Defender status...
[2026-01-06 14:32:16] [INFO] Initial Status - Service Running: True
[2026-01-06 14:32:16] [INFO] Initial Status - Real-Time Protection: False
[2026-01-06 14:32:16] [INFO] Initial Status - Antivirus Enabled: True
[2026-01-06 14:32:16] [INFO] Checking for third-party antivirus software...
[2026-01-06 14:32:17] [WARNING] Third-party AV detected: Avast Free Antivirus
[2026-01-06 14:32:17] [INFO] Diagnostic check completed. No changes made.
[2026-01-06 14:32:20] [INFO] Checking Microsoft Defender status...
[2026-01-06 14:32:21] [INFO] === Final Status ===
[2026-01-06 14:32:21] [INFO] Service Running: True
[2026-01-06 14:32:21] [INFO] Real-Time Protection: False
[2026-01-06 14:32:21] [INFO] Antivirus Enabled: True
[2026-01-06 14:32:21] [WARNING] ⚠ Microsoft Defender may require additional
action or system reboot
[2026-01-06 14:32:21] [INFO] === Script Completed ===
```

End User Experience:

Metric	Value
Duration	6 seconds
Screen Impact	None
Performance Impact	Minimal CPU for WMI queries
Interruption Level	**Zero**
User Notifications	None
Service Interruption	No

Outcome: ☑ Identified issue (third-party AV blocking Defender) without making any changes.

Recommended Workflow

```
┌──────────────────────────────┐
│  Start: Defender Issue       │
│  Reported or Detected        │
└──────────────────────────────┘
           │
           ▼
┌──────────────────────────────┐
│  Step 1: Diagnostic Check    │
│  (0 user impact)             │
└──────────────────────────────┘
           │
           ▼
     ┌──────────┐
     │  Healthy?  │
     └──────────┘
       │       │
      Yes      No
       │       │
       │       ▼
       │   ┌──────────────────────────┐
       │   │  Step 2: Full Repair     │
       │   │  (Smart - minimal impact)│
       │   └──────────────────────────┘
       │              │
       │              ▼
       │        ┌──────────┐
       │        │  Fixed?    │
       │        └──────────┘
       │          │       │
       │         Yes      No
       │          │       │
       │          │       ▼
       │          │   ┌──────────────────────────┐
       │          │   │  Step 3: Force Full Repair  │
       │          │   │  (Nuclear - moderate impact)│
       │          │   └──────────────────────────┘
       │          │              │
       │          │              ▼
       │          │        ┌──────────┐
       │          │        │  Fixed?    │
       │          │        └──────────┘
       │          │          │       │
       │          │         Yes      No
       │          │          │       │
       │          │          │       ▼
       │          │          │   ┌──────────────────────────┐
       │          │          │   │  Step 4: Escalate        │
       │          │          │   │  - DISM/SFC              │
       │          │          │   │  - Windows Repair        │
       │          │          │   │  - Manual intervention   │
```

```
        │          │          │   └──────────────────────────┘
        │          │          │
        ▼          ▼          ▼
  ┌──────────────────────────┐
  │  Document & Close Ticket  │
  └──────────────────────────┘
```

NinjaRMM Configuration

Custom Fields Required

Field Name: `defenderRepairStatus`
Type: Text/Wysiwyg
Purpose: Stores last repair action and result

Possible Values:

- `Success: Diagnostic Check Only`
- `Success: Enable Defender`
- `Success: Reset to Default`
- `Success: Full Repair (All Steps)`
- `Success: Force Full Repair (No Checks)`
- `Failed: [Action Name]`

Script Parameters

Parameter Name: `repairAction`
Type: Dropdown
Values:

- Diagnostic Check Only (default)
- Enable Defender
- Reset to Default
- Remove Third-Party AV
- Full Repair (All Steps)
- Force Full Repair (No Checks)

Deployment Settings

- **Run As:** Administrator (required)
- **Script Type:** PowerShell
- **Timeout:** 120 seconds
- **Run Condition:** Manual or scheduled
- **Notification:** On failure only

Impact Summary by Option

Low Impact (Recommended First)

Diagnostic Check Only

- Duration: 6-8 seconds
- User awareness: 0%
- Protection gap: None
- Risk level: None

Full Repair (Smart)

- Duration: 8-20 seconds (varies)
- User awareness: 0-5%
- Protection gap: 0-5 seconds (if needed)
- Risk level: Very Low

Moderate Impact (Escalation)

Enable Defender

- Duration: 5-10 seconds
- User awareness: 5-10%
- Protection gap: 3-5 seconds
- Risk level: Low

Reset to Default

- Duration: 10-15 seconds
- User awareness: 10-15%
- Protection gap: 5-10 seconds
- Risk level: Low-Moderate

Higher Impact (Last Resort)

Force Full Repair

- Duration: 18-25 seconds
- User awareness: 15-20%
- Protection gap: 5-8 seconds
- Risk level: Moderate

Common Issues & Resolutions

Symptom	Likely Cause	Recommended Action
Real-Time Protection off	Third-party AV conflict	Full Repair (Smart)
Service not running	Manual disable or policy	Enable Defender
Policy overrides present	GPO or malware	Reset to Default
PowerShell cmdlets missing	Windows corruption	Force Repair → DISM/SFC

Symptom	Likely Cause	Recommended Action
Appears healthy but isn't	Subtle corruption	Force Full Repair
Third-party AV detected	Recent uninstall incomplete	Remove Third-Party AV → Full Repair

---

Success Metrics

Script Success Rates (Expected)

- **Diagnostic Check:** 100% (read-only)
- **Enable Defender:** 85-90%
- **Reset to Default:** 80-85%
- **Full Repair (Smart):** 75-85%
- **Force Full Repair:** 60-70%

When Scripts Fail

If Force Full Repair fails, the issue is likely:

1. **Windows component corruption** (requires DISM/SFC)
2. **Hardware/firmware TPM issues**
3. **Deep malware infection**
4. **Windows installation requires repair/reset**

Security Considerations

Protection Gaps

- **Diagnostic Check:** No gap
- **Enable Only:** 3-5 seconds
- **Reset:** 5-10 seconds
- **Full/Force Repair:** 5-8 seconds

Risk Mitigation

1. Schedule during maintenance windows when possible
2. Ensure third-party AV is removed before repair
3. Verify system is not actively under attack
4. Have rollback plan (though script is non-destructive)

Troubleshooting Guide

Script Exits with Code 1

Possible Causes:

- PowerShell module missing

- Permissions insufficient
- Service cannot start
- Registry access denied

Resolution:

1. Verify admin rights
2. Check Windows Update
3. Run DISM/SFC
4. Review Event Viewer logs

No Changes Made (Smart Repair)

This is by design when:

- System is already healthy
- No policy overrides found
- All services running properly

Next Steps:

- If issue persists, use Force Full Repair
- Check for hidden malware
- Review application event logs

Third-Party AV Detected

Script Action:

- Logs AV presence
- Shows uninstall string
- Recommends manual removal

Manual Steps:

1. Use vendor removal tool
2. Reboot system
3. Run script again

Support Contact

Script Issues:

- Review NinjaRMM Activity Log
- Check `defenderRepairStatus` custom field
- Examine full console output

Escalation Path:

1. Try next repair level
2. DISM/SFC commands

3. Windows Repair
4. Microsoft Support

Version History

v1.1 - January 6, 2026

- Added Force Full Repair option
- Implemented intelligent repair logic
- Enhanced logging detail
- Improved third-party AV detection

v1.0 - Initial Release

- Basic repair options
- NinjaRMM integration
- Status diagnostics

Appendix: Technical Details

Registry Keys Modified

```
HKLM:\SOFTWARE\Policies\Microsoft\Windows Defender
HKLM:\SOFTWARE\Policies\Microsoft\Windows Defender\Real-Time Protection
HKLM:\SOFTWARE\Policies\Microsoft\Windows Advanced Threat Protection
```

Services Affected

- **WinDefend** (Windows Defender Service)
- Startup Type set to: Automatic
- Service action: Start/Restart as needed

PowerShell Cmdlets Used

- `Get-MpComputerStatus`
- `Get-MpPreference`
- `Set-MpPreference`
- `Get-Service`
- `Set-Service`
- `Start-Service`
- `Restart-Service`
- `Get-CimInstance`

WMI Namespaces Queried

- `root/SecurityCenter2` (AntiVirusProduct detection)

Document End

This simulation document represents expected behavior based on common scenarios. Actual results may vary based on system configuration, Windows version, and specific issues present.